PHYS 143 – Mechanics and Thermal Physics

Week 1: Vectors – Tutorial Sheet

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Use the Master Procedure: identify, sketch, choose axes, predict, resolve, combine, reconstruct, evaluate. Give directions as angles measured counter-clockwise from $+x$ unless stated. Final answers to the practice problems and to the exam-style items are at the end.

A. Practice problems

A1. Magnitude and direction

Find the magnitude and the direction, measured counter-clockwise from $+x$, of $\vec{\boldsymbol{A}} = (-6.0\hat{\boldsymbol{\imath}} + 8.0\hat{\boldsymbol{\jmath}})\,\mathrm{m}$. Sketch first and name the quadrant.

A2. Two displacements

A walker goes $3.0\,\mathrm{km}$ east and then $4.0\,\mathrm{km}$ at 30° north of west. Find the resultant displacement (magnitude and direction from $+x$) using components.

A3. Dot and cross products

For $\vec{\boldsymbol{A}} = (2.0\hat{\boldsymbol{\imath}} + 3.0\hat{\boldsymbol{\jmath}} - 1.0\hat{\boldsymbol{k}})$ and $\vec{\boldsymbol{B}} = (1.0\hat{\boldsymbol{\imath}} - 2.0\hat{\boldsymbol{\jmath}} + 4.0\hat{\boldsymbol{k}})$ find $\vec{\boldsymbol{A}} \cdot \vec{\boldsymbol{B}}$, $\vec{\boldsymbol{A}} \times \vec{\boldsymbol{B}}$ and the angle between the vectors. Predict first whether the angle is acute or obtuse.

A4. Torque

A force of $60\,\mathrm{N}$ acts at the end of a $0.35\,\mathrm{m}$ spanner. The angle between the spanner and the force is 40°. Find the magnitude of the torque about the pivot.

A5. Work

A $50\,\mathrm{N}$ force at 25° above the horizontal drags a crate $12\,\mathrm{m}$ along level concrete. Find the work done by the force.

B. Extra worked examples (moved from the lecture)

B1. Four displacements (the delivery rider)

Given: legs of $2.1\,\text{km}$ at $20°$ N of E, $3.4\,\text{km}$ at $35°$ W of N, $1.8\,\text{km}$ S, $2.6\,\text{km}$ at $10°$ S of W. Find the final displacement.
Sketch: four legs head to tail; the rider ends up north-west of the start (predict $R_x < 0$, $R_y > 0$).
IDENTIFY: resultant $\vec{R}$ of four vectors: resolve, combine, reconstruct.
SET UP: $+x$ east, $+y$ north; Rule (6) with signs read from the sketch; $R_x = \sum A_x$, $R_y = \sum A_y$.
EXECUTE:
x-components: $+2.1\cos 20° = +1.97$, $-3.4\sin 35° = -1.95$, 0, $-2.6\cos 10° = -2.56$ (km).
y-components: $+2.1\sin 20° = +0.72$, $+3.4\cos 35° = +2.79$, -1.80, $-2.6\sin 10° = -0.45$ (km).
Substitute: $R_x = 1.97 - 1.95 + 0 - 2.56 = -2.54\,\text{km}$.
Substitute: $R_y = 0.72 + 2.79 - 1.80 - 0.45 = +1.25\,\text{km}$.
Substitute: $R = \sqrt{(-2.54\,\text{km})^2 + (1.25\,\text{km})^2} = 2.83\,\text{km}$.
Reference angle $\tan^{-1}(1.25/2.54) = 26.2°$; quadrant II, so $\theta = 180° - 26.2° = 153.7°$ from $+x$.
$\boxed{R = 2.83\,\text{km} \text{ at } 153.7° \text{ from } +x \text{ (about } 26° \text{ north of west)}}$
EVALUATE: quadrant II as predicted; $R = 2.83 <$ sum of the legs ($9.9\,\text{km}$); units km.

B2. Angle between two vectors in components

Given: $\vec{A} = (2\hat{\imath} + 3\hat{\jmath})\,\text{m}$, $\vec{B} = (4\hat{\imath} + \hat{\jmath})\,\text{m}$. Find the angle between them.
Sketch: both vectors lie in quadrant I with A steeper; predict ϕ small and acute.
IDENTIFY: angle from the dot product.
SET UP: Rules (12) and (11): $\cos\phi = \dfrac{\vec{A}\cdot\vec{B}}{AB}$.
EXECUTE:
Substitute: $\vec{A}\cdot\vec{B} = (2)(4) + (3)(1) = 11\,\text{m}^2$.
Substitute: $A = \sqrt{2^2 + 3^2} = 3.61\,\text{m}$, $B = \sqrt{4^2 + 1^2} = 4.12\,\text{m}$.
Substitute: $\cos\phi = \dfrac{11}{(3.61)(4.12)} = 0.740$.
$\boxed{\phi = \cos^{-1}(0.740) = 42.3°}$
EVALUATE: acute, as the positive dot product predicts; the directions $56.3°$ and $14.0°$ differ by $42.3°$. ✓

Worked solutions to A1 to A5

A1. Magnitude and direction

Given: $\vec{A} = (-6.0\hat{\imath} + 8.0\hat{\jmath})\,\text{m}$. **Sketch:** left and up, quadrant II.
IDENTIFY and SET UP: reconstruct with Rule (7); quadrant from the signs.
EXECUTE:
Substitute: $A = \sqrt{(-6.0)^2 + (8.0)^2}\,\text{m}$, $\boxed{A = 10.0\,\text{m}}$
Substitute: $\theta_{\text{ref}} = \tan^{-1}(8.0/6.0) = 53.1°$.
Quadrant II: $\theta = 180° - 53.1° = \boxed{126.9°}$
EVALUATE: the calculator gives $-53.1°$, which points right and down, so it is rejected; $126.9°$ agrees with the sketch.

A2. Two displacements

Given: $3.0\,\text{km}$ east, $4.0\,\text{km}$ at $30°$ north of west. **Sketch:** second leg goes left and up; predict R_x small, $R_y > 0$.
SET UP: $A_x = +3.0$, $A_y = 0$; $B_x = -4.0\cos 30°$, $B_y = +4.0\sin 30°$; Rule (8).
EXECUTE:
Substitute: $R_x = 3.0 - 4.0\cos 30° = 3.0 - 3.46 = \boxed{-0.46\,\text{km}}$
Substitute: $R_y = 0 + 4.0\sin 30° = \boxed{+2.00\,\text{km}}$
Substitute: $R = \sqrt{(-0.46)^2 + (2.00)^2}\,\text{km} = \boxed{2.05\,\text{km}}$
Reference angle $\tan^{-1}(2.00/0.46) = 77.0°$; quadrant II, so $\theta = 180° - 77.0° = \boxed{103°\text{ from } +x}$
EVALUATE: quadrant II as predicted; $2.05 < 7.0\,\text{km}$.

A3. Dot and cross products

Given: $\vec{A} = (2\hat{\imath} + 3\hat{\jmath} - \hat{k})$, $\vec{B} = (\hat{\imath} - 2\hat{\jmath} + 4\hat{k})$. **Predict:** $\vec{A}\cdot\vec{B}$ is likely negative, so ϕ obtuse.
SET UP: Rules (12), (14), (9), (11).
EXECUTE:
Substitute: $\vec{A}\cdot\vec{B} = (2)(1) + (3)(-2) + (-1)(4) = 2 - 6 - 4 = \boxed{-8}$
$\vec{A}\times\vec{B}$: $\hat{\imath}$: $(3)(4) - (-1)(-2) = 10$; $\hat{\jmath}$: $(-1)(1) - (2)(4) = -9$; $\hat{k}$: $(2)(-2) - (3)(1) = -7$.
$\boxed{\vec{A}\times\vec{B} = 10\hat{\imath} - 9\hat{\jmath} - 7\hat{k}}$, magnitude $\sqrt{100 + 81 + 49} = 15.17$.
Substitute: $\cos\phi = \dfrac{-8}{\sqrt{14}\sqrt{21}} = -0.467$, $\boxed{\phi = 117.8°}$
EVALUATE: obtuse, as the negative dot product predicts; $(\vec{A}\times\vec{B})\cdot\vec{A} = 20 - 27 + 7 = 0$ ✓; $15.17 = AB\sin\phi$.

A4. Torque

Given: $F = 60\,\text{N}$, $r = 0.35\,\text{m}$, $\phi = 40°$. **Predict:** $\tau < rF = 21\,\text{N m}$.
SET UP: Rule (13): $\tau = rF\sin\phi$.
Substitute: $\tau = (0.35\,\text{m})(60\,\text{N})\sin 40° = (21.0)(0.643)\,\text{N m}$
$\boxed{\tau = 13.5\,\text{N m}}$
EVALUATE: less than $21\,\text{N m}$ as predicted; units N m.

A5. Work

Given: $F = 50\,\text{N}$, $\phi = 25°$ from the displacement, $s = 12\,\text{m}$. **Predict:** $W < Fs = 600\,\text{J}$, positive.
SET UP: Rule (11): $W = \vec{F} \cdot \vec{s} = Fs\cos\phi$.
Substitute: $W = (50\,\text{N})(12\,\text{m})\cos 25° = (600)(0.9063)\,\text{J}$
$\boxed{W = 544\,\text{J}}$
EVALUATE: positive and below $600\,\text{J}$, as predicted; only the component along the motion does work.

Exam-style practice

Ten five-option items in the style of the course examination, **suggested time: 3 minutes per item**. For each item sketch first, predict the sign or size, then calculate (IDENTIFY, SET UP, EXECUTE, EVALUATE). Choose one answer, A to E. Items marked as sharing information are linked. Answers and brief solutions are at the end of this sheet. Take $g = 9.80\,\text{m/s}^2$.

E1. A displacement $\vec{D}$ of magnitude $20.0\,\text{m}$ points $30°$ south of west. Its x and y components ($+x$ east, $+y$ north), in metres, are

(A) $(-17.3,\ -10.0)$

(B) $(-17.3,\ 10.0)$

(C) $(-10.0,\ -17.3)$

(D) $(10.0,\ 17.3)$

(E) $(17.3,\ -10.0)$

E2. $\vec{A} = (3.0\hat{\imath} - 5.0\hat{\jmath})\,\text{m}$ and $\vec{B} = (-7.0\hat{\imath} + 2.0\hat{\jmath})\,\text{m}$. The direction of $\vec{A} + \vec{B}$, measured counter-clockwise from $+x$, is

(A) $36.9°$ **(B)** $126.9°$ **(C)** $143.1°$ **(D)** $216.9°$ **(E)** $323.1°$

E3. Which of the following statements is/are true for two non-zero vectors $\vec{A}$ and $\vec{B}$?

I. $|\vec{A} + \vec{B}| = |\vec{A}| + |\vec{B}|$ only if $\vec{A}$ and $\vec{B}$ are parallel.
II. $\vec{A} \times \vec{B} = \vec{B} \times \vec{A}$.
III. $\vec{A} \cdot \vec{B}$ is a scalar and can be negative.

(A) I only **(B)** I and II only **(C)** I and III only **(D)** II and III only **(E)** I, II and III

E4. For $\vec{P} = (\hat{\imath} - 3\hat{\jmath} + 2\hat{k})$ and $\vec{Q} = (2\hat{\imath} + \hat{\jmath} - \hat{k})$, the magnitude of $\vec{P} \times \vec{Q}$ is

(A) 3.00 **(B)** 7.35 **(C)** 8.66 **(D)** 9.17 **(E)** 13.0

Items E5 and E6 share this information: A force $\vec{F} = (4.0\hat{\imath} - 2.0\hat{\jmath} + 3.0\hat{k})\,\text{N}$ acts on a body that undergoes the displacement $\vec{d} = (2.0\hat{\imath} + 5.0\hat{\jmath} - 1.0\hat{k})\,\text{m}$.

E5. The work done by $\vec{F}$ during the displacement is

(A) $-18.0\,\text{J}$ **(B)** $-5.0\,\text{J}$ **(C)** $5.0\,\text{J}$ **(D)** $18.0\,\text{J}$ **(E)** $29.5\,\text{J}$

E6. Using your previous result, the angle between $\vec{F}$ and $\vec{d}$ is closest to

(A) 45° **(B)** 80° **(C)** 90° **(D)** 100° **(E)** 170°

E7. Three forces act at one point: $50.0\,\mathrm{N}$ along $+x$, $30.0\,\mathrm{N}$ at 120° from $+x$, and $40.0\,\mathrm{N}$ along $-y$. The magnitude of the resultant is

(A) $19.0\,\mathrm{N}$ **(B)** $37.7\,\mathrm{N}$ **(C)** $52.0\,\mathrm{N}$ **(D)** $70.7\,\mathrm{N}$ **(E)** $120.0\,\mathrm{N}$

E8. A force of magnitude $15\,\mathrm{N}$ is directed along $\vec{A} = (2\hat{\imath} - \hat{\jmath} + 2\hat{k})$. The force, in newtons, is

(A) $2\hat{\imath} - 1\hat{\jmath} + 2\hat{k}$

(B) $5\hat{\imath} - 5\hat{\jmath} + 5\hat{k}$

(C) $10\hat{\imath} - 5\hat{\jmath} + 10\hat{k}$

(D) $10\hat{\imath} + 5\hat{\jmath} + 10\hat{k}$

(E) $30\hat{\imath} - 15\hat{\jmath} + 30\hat{k}$

E9. A force of $25.0\,\mathrm{N}$ acts at the end of a $0.4\,\mathrm{m}$ spanner. The angle between the spanner (tail to tail with the force) and the force is 150°. The magnitude of the torque $\vec{r} \times \vec{F}$ about the pivot is

(A) $2.5\,\mathrm{N\,m}$

(B) $5.0\,\mathrm{N\,m}$

(C) $8.7\,\mathrm{N\,m}$

(D) $10.0\,\mathrm{N\,m}$

(E) $20.0\,\mathrm{N\,m}$

E10. Vectors $\vec{A}$ and $\vec{B}$ have magnitudes $5.0\,\mathrm{m}$ and $3.0\,\mathrm{m}$. Which value of $|\vec{A} + \vec{B}|$ is impossible?

(A) $2\,\mathrm{m}$ **(B)** $4\,\mathrm{m}$ **(C)** $6\,\mathrm{m}$ **(D)** $8\,\mathrm{m}$ **(E)** $10\,\mathrm{m}$

Answers to the practice problems

A1 $|\vec{A}| = 10.0$; $\theta = 126.9^\circ$ (quadrant II, not -53.1°)
A2 $R_x = -0.46$, $R_y = 2.00$ (km), $R = 2.05\,\mathrm{km}$ at 103° from $+x$ (quadrant II)
A3 $\vec{A} \cdot \vec{B} = -8$; $\vec{A} \times \vec{B} = 10\hat{\imath} - 9\hat{\jmath} - 7\hat{k}$, magnitude 15.17; $\phi = 117.8^\circ$
A4 $\tau = rF\sin\phi = 13.5\,\mathrm{N\,m}$
A5 $W = Fs\cos\phi = 544\,\mathrm{J}$

Exam-style practice: answer key and brief solutions

Key: E1 A E2 D E3 C E4 C E5 B E6 D E7 B E8 C E9 B E10 E

E1 (A). Sketch first: south of west lies in quadrant III, so both components are negative. The 30° is measured from the west axis, so $D_x = -D\cos 30^\circ = -17.3$ and $D_y = -D\sin 30^\circ = -10.0$. Check: $\sqrt{D_x^2 + D_y^2} = 20.0$.

E2 (D). $\vec{R} = (-4.0\hat{\imath} - 3.0\hat{\jmath})\,\mathrm{m}$: both components are negative, so $\vec{R}$ is in quadrant III (sketch). Reference angle $\tan^{-1}(3.0/4.0) = 36.9^\circ$, so $\theta = 180^\circ + 36.9^\circ = 216.9^\circ$. The calculator value 36.9° alone is the quadrant I trap.

E3 (C). I is true (the magnitudes add only when the vectors point the same way). II is false: $\vec{\boldsymbol{A}} \times \vec{\boldsymbol{B}} = -\vec{\boldsymbol{B}} \times \vec{\boldsymbol{A}}$, so the order matters. III is true: $\vec{\boldsymbol{A}} \cdot \vec{\boldsymbol{B}} = AB\cos\phi$ is negative for $\phi > 90°$. Answer: I and III only.

E4 (C). $\vec{\boldsymbol{P}} \times \vec{\boldsymbol{Q}} = \big[(-3)(-1) - (2)(1)\big]\hat{\boldsymbol{\imath}} - \big[(1)(-1) - (2)(2)\big]\hat{\boldsymbol{\jmath}} + \big[(1)(1) - (-3)(2)\big]\hat{\boldsymbol{k}} = \hat{\boldsymbol{\imath}} + 5\hat{\boldsymbol{\jmath}} + 7\hat{\boldsymbol{k}}$, so the magnitude is $\sqrt{1 + 25 + 49} = 8.66$. Check: it cannot exceed $PQ = 9.17$. The value $|PQ| = 9.17$ would need $\vec{\boldsymbol{P}} \perp \vec{\boldsymbol{Q}}$, but here $\vec{\boldsymbol{P}} \cdot \vec{\boldsymbol{Q}} = -3 \neq 0$.

E5 (B). $\vec{\boldsymbol{F}} \cdot \vec{\boldsymbol{d}} = (4.0)(2.0) + (-2.0)(5.0) + (3.0)(-1.0) = 8.0 - 10.0 - 3.0 = -5.0\,\mathrm{J}$. The negative sign means that, on balance, $\vec{\boldsymbol{F}}$ has a component against the displacement.

E6 (D). $\cos\phi = \dfrac{\vec{\boldsymbol{F}} \cdot \vec{\boldsymbol{d}}}{Fd} = \dfrac{-5.0}{(5.385)(5.477)} = -0.170$, so $\phi = 99.8°$: slightly obtuse, as the negative dot product predicted.

E7 (B). $R_x = 50.0 + 30.0\cos 120° = 35.0\,\mathrm{N}$, $R_y = 30.0\sin 120° - 40.0 = -14.0\,\mathrm{N}$, so $R = \sqrt{35.0^2 + 14.0^2} = 37.7\,\mathrm{N}$, pointing right and down (quadrant IV). Adding the magnitudes ($120\,\mathrm{N}$) is the usual error.

E8 (C). $|\vec{\boldsymbol{A}}| = \sqrt{4 + 1 + 4} = 3$, so $\hat{\boldsymbol{A}} = \frac{1}{3}(2\hat{\boldsymbol{\imath}} - \hat{\boldsymbol{\jmath}} + 2\hat{\boldsymbol{k}})$ and $\vec{\boldsymbol{F}} = 15\,\hat{\boldsymbol{A}} = 10\hat{\boldsymbol{\imath}} - 5\hat{\boldsymbol{\jmath}} + 10\hat{\boldsymbol{k}}$. Check: $\sqrt{100 + 25 + 100} = 15$. The vector $\vec{\boldsymbol{A}}$ itself has magnitude 3, not 15.

E9 (B). $\tau = rF\sin\phi = (0.40)(25.0)\sin 150° = 5.0\,\mathrm{N\,m}$. The $150°$ is the angle between the vectors, and $\sin 150° = \sin 30°$, so only half of rF counts. Using $\cos$ instead of $\sin$ gives $8.7\,\mathrm{N\,m}$, and omitting the angle gives $10.0\,\mathrm{N\,m}$.

E10 (E). $|\vec{\boldsymbol{A}} + \vec{\boldsymbol{B}}|$ lies between $|A - B| = 2.0\,\mathrm{m}$ (antiparallel) and $A + B = 8.0\,\mathrm{m}$ (parallel). So $10\,\mathrm{m}$ is impossible; the others are all attainable.